

Atul Anurag, Ph.D.

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EDUCATION

New Jersey Institute of Technology (NJIT), USA

Sept 2019 – July 2025

Ph.D. in Applied Mathematics

Dissertation Advisor: [Roy H. Goodman](#)

National Institute of Technology (NIT), Warangal, India

Aug 2015 – Jun 2017

M.Sc. in Applied Mathematics

Dissertation Advisor: [Poosan Muthu Moopanar](#)

Ramjas College, University of Delhi (DU), New Delhi

Jul 2012 – Jul 2015

B.Sc. (Honors) in Pure Mathematics

RESEARCH PROFILE

Applied mathematician specializing in nonlinear dynamical systems, scientific computing, and mathematical modeling of complex physical systems. My research combines analytical and computational methods to study stability, bifurcations, and long-term dynamics in Hamiltonian and fluid systems. Recent work includes the development of novel reduction techniques, phase-space analysis, and numerical methods for vortex dynamics and related nonlinear phenomena.

Research Interests: Nonlinear Dynamical Systems, Scientific Computing, Mathematical Modeling, Computational Fluid Dynamics, Hamiltonian Systems, Bifurcation Theory, Numerical Analysis, Complex Systems.

EXPERIENCE

Assistant (Teaching) Professor of Mathematics

Lander University, Greenwood, South Carolina

August 2026 – Present

Adjunct Professor of Mathematics

Ramapo College of New Jersey, Mahwah, New Jersey

July 2026 – August 2026

- Teach Calculus II and Precalculus during the Summer II term.

Assistant Professor of Mathematics (Visiting)

Ramapo College of New Jersey, Mahwah, New Jersey

Sep 2025 – June 2026

- Independently teach undergraduate courses including Precalculus, Differential Equations, Probability and Statistics, and Mathematical Modeling.
- Teach multiple sections per semester, adapting instruction to students with diverse academic backgrounds.
- Design course materials, assessments, and structured lessons to improve conceptual understanding and performance.
- Provide targeted academic support through office hours and review sessions to strengthen foundational skills.

Research Assistant

Under the supervision of Dr. Roy H. Goodman, NJIT

Spring 2022 – Summer 2025

- Developed MATLAB algorithms to analyze the global phase space of a three-point vortex problem on a sphere using a novel reduction technique.
- Designed computational methods for the generalization of leapfrogging orbits of point vortices using [AUTO](#). My talk can be found [here](#).
- Developed a new coordinate system to handle zero total vortex strength, overcoming conventional limitations.

Teaching Assistant & Adjunct Professor

New Jersey Institute of Technology

Fall 2019 – Spring 2022

- Adjunct professor for calculus courses at NJIT's [Educational Opportunity Program Summer Program](#).
- Led recitations and instructional support for Calculus I & II and MATLAB programming. Notes available [here](#).
- Assisted students with problem-solving and computational techniques.

Research Co-Mentor, NJIT

Summer 2023 – Summer 2025

- Co-mentored undergraduate researchers on mathematical modeling and computational projects; supervised E. O'Grady's research, resulting in a peer-reviewed publication.

Summer Researcher

Indraprastha Institute of Information Technology, New Delhi, India

Summer 2018

- Applied operator theory to analyze nonlinear PDEs.
- Advisor: [Prof. Ashish Kumar Pandey](#).

Indian Academy of Sciences Intern

Tata Institute of Fundamental Research, Bangalore, India

Summer 2014

- Implemented image and digital signal processing algorithms using Laplace transforms.
- Advisor: [Prof. Kayyannapara Thomas Joseph](#).

PUBLICATIONS

- **Atul Anurag** and Roy H. Goodman, *Phase Portraits and the Bifurcation Set for the Three-Vortex Interaction System*. *Communications in Nonlinear Science and Numerical Simulation*, 2026. [\[Link\]](#)
- **Atul Anurag**, Roy H. Goodman, and Ellison O'Grady, "A New Canonical Reduction of Three-Vortex Motion and Its Application to Vortex-Dipole Scattering," *Physics of Fluids*, vol. 36, no. 6, 067110, 2024. [\[Link\]](#)
- **Atul Anurag** and Roy H. Goodman, *The Four-Vortex Motion with Zero Total Circulation*. In preparation.
- **Atul Anurag**, *The Global Phase Space of the Three-Vortex Interaction System and Its Application to Vortex-Dipole Scattering*. PhD Dissertation, New Jersey Institute of Technology, 2025. [\[Link\]](#)

SELECTED GRADUATE COURSEWORK

Asymptotic Methods, Optimization Theory, Applied Statistics, Computational Fluid Dynamics, Discrete Mathematics, Functional Analysis, Finite Volume Methods, Topology, Integral and Discrete Transforms, Finite Element Methods, Theoretical Operations Research, Number Theory, etc.

CONFERENCE AND SEMINAR PRESENTATIONS

- (Poster) [The Global Phase Plane Analysis of Three Vortex Interactions](#), Frontiers in Applied & Computational Mathematics, NJIT, June 2025.
- (Joint work with Roy Goodman) [The Phase Space of the Three-Vortex Problem and Its Application to Vortex-Dipole Scattering](#), SIAM DS25, Denver, May 2025.
- (Poster) [The Global Phase Plane Analysis of Three Vortex Interactions](#), Board Day and Dana Knox Research Showcase, NJIT, April 2025.
- (Poster) [Global Phase Plane Analysis of the three-vortex problem](#), SIAM-NNP, Rochester Institute of Technology, November 2024.
- (Talk) [The Phase Space of the Three-Vortex Problem](#), NJIT, June 2024.
- (Talk) [The Phase Space of the Three-Vortex Problem](#), SIAM Nonlinear Waves Conference, Baltimore, June 2024.

- (Poster) [Point Vortex Dipole Scattering](#), SIAM New York – New Jersey – Pennsylvania (SIAM-NNP), NJIT, October (2023).
- (Talk) Continuation of Periodic Orbits in Symmetric Hamiltonian and Conservative Systems, Faculty and Student Summer Talks, Mathematics, NJIT, July (2023).
- (Problem Solver) Mathematical Problems in Industry Workshop, NJIT, June (2023).
- (Attendee) Second Drexel Waves Workshop, Drexel University, March (2023).
- (Thesis Proposal Defense) Generalization of Leapfrogging Orbits of Point Vortices, NJIT, January (2023).
- (Talk) Walking Droplet Dynamics Research, NJIT, June (2021).

SKILLS

Programming Languages: Python, MATLAB, Mathematica, Julia, FORTRAN, C++, HTML, R, SQL

Other Software: $\text{\LaTeX}$, [AUTO](#)

Languages: Fluent in Hindi, English, and Sanskrit

AWARDS AND HONORS

- Research Fellowship at NJIT, supported by the NSF under [DMS-2206016](#) Oct 2022 - Aug 2025
- Teaching Fellowship at NJIT Sep 2019 - Oct 2022
- [CSIR Junior Research Fellowship](#), All India Rank 46 among 40,000 students
Only top 200 students are fully funded in their PhD program 2018
- [Indian Institute of Technology, Joint Admission Test for M.Sc. \(IIT-JAM\)](#)
Secured All India Rank 354 among 15,000 (approx) students 2015
- [Indian Academy of Sciences Fellowship](#) Summer 2014

LEADERSHIP AND SERVICE

- [Student Chapter of SIAM at NJIT](#), Vice-President June 2022 – June 2024
- [UCAN Executive Committee](#), Grad executive board member-at-large,
A Union of Student Workers, Researchers, and Adjunct Instructors June 2024 – June 2025
- (Volunteer, Attendee) Frontiers in Applied and Computational Mathematics, NJIT 2022, 2023
- Class Representative (M.Sc.), Department of Mathematics, NIT, Warangal, India July 2015 – May 2017

REFERENCES

Dr. Roy H. Goodman, Department of Mathematical Sciences, NJIT
Dissertation Advisor, goodman@njit.edu

Dr. David Shirokoff, Department of Mathematical Sciences, NJIT
Dissertation Committee Member, david.shirokoff@njit.edu

Dr. Anand Oza, Department of Mathematical Sciences, NJIT
Academic Reference, oja@njit.edu

Dr. Katarzyna Kowal, Department of Mathematical Sciences, Ramapo College of New Jersey
Teaching Reference, kkowal@ramapo.edu